

LU HONGYI

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Education

HKUST Ph.D. Computer Science	2022 – 2026
SUSTech B.Sc. Mathematics	2017 – 2021

Working Experience

NIO SkyOS Team <i>Software Expert</i> • Design seL4-based system software (VMM/OS) to supports AI applications.	2026 – Present
SUSTech <i>Research Assistant</i> • Conducted research in computer systems.	2021 – 2022

Publications

[1] Lijian Huang*, **Hongyi Lu***, Shuai Wang, and Fengwei Zhang. Towards Secure BPF Kernel Extension with Hardware-enhanced Memory Isolation. In *the Transactions on Dependable and Secure Computing (TDSC, CCF-A)*, 2026.

[2] Zihao Li*, **Hongyi Lu***, Yanan Guo, Zhenkai Zhang, Shuai Wang, and Fengwei Zhang. GPU-Fuzz: Finding memory errors in deep learning frameworks. In *the Proceedings of the 63th ACM/IEEE Design Automation Conference (DAC, CCF-A)*, 2026.

[3] **Hongyi Lu**, Fengwei Zhang, Zhenkai Zhang, Shuai Wang, and Yanan Guo. CuSAFE: Capturing Memory Corruption on NVIDIA GPUs. In *the 35nd USENIX Security Symposium (SEC, CCF-A)*, 2026.

[4] **Hongyi Lu**. Hardware-assisted Memory Isolation. In *the Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS, PhD Symposium)*, 2025.

[5] **Hongyi Lu***, Yunjie Deng*, Sukarno Mertoguno, Shuai Wang, and Fengwei Zhang. MOLE: Breaking GPU TEE with GPU-Embedded MCU. In *the Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security (CCS, CCF-A)*, 2025.

[6] **Hongyi Lu**, Zhibo Liu, Shuai Wang, and Fengwei Zhang. DTD: Comprehensive and Scalable Testing for Debuggers. In *the Proceedings of the ACM on Software Engineering (FSE, CCF-A)*, 2024.

[7] **Hongyi Lu**, Shuai Wang, Yechang Wu, Wanning He, and Fengwei Zhang. MOAT: Towards Safe BPF Kernel Extension. In *33nd USENIX Security Symposium (USENIX Security, CCF-A)*, 2024.

[8] Wanning He*, **Hongyi Lu***, Fengwei Zhang, and Shuai Wang. RingGuard: Guard io_uring with eBPF. In *the Proceedings of the 1st Workshop on eBPF and Kernel Extensions (eBPF)*, pages 56–62, 2023.

[9] Haonan Li, Weijie Huang, Mingde Ren, **Hongyi Lu**, Zhenyu Ning, Heming Cui, and Fengwei Zhang. A Novel Memory Management for RISC-V Enclaves. In *the Proceedings of the 10th International Workshop on Hardware and Architectural Support for Security and Privacy (HASP)*, 2022.

[10] **Hongyi Lu** and Fengwei Zhang. Raven: a Novel Kernel Debugging Tool on RISC-V. In *the Proceedings of the 59th ACM/IEEE Design Automation Conference (DAC, CCF-A)*, pages 1039–1044, 2022.

[11] **Hongyi Lu**, Yechang Wu, Shuqing Li, You Lin, Chaozu Zhang, and Fengwei Zhang. BadUSB-C: Revisiting BadUSB with Type-C. In *2021 IEEE Security and Privacy Workshops (WOOT)*, pages 327–338, 2021.

In total, I have published **seven CCF-A papers** as first/co-first author (marked with *), **three in Security Top 4 conferences**.

Academic Service

Official Reviewer	TIFS, TDSC
Ext. Reviewer	USENIX Security 2023, S&P 2023, PoPET 2023, CCS 2023, CCS 2022

Grant & Awards

CCS Student Grant \$1750	2025
USENIX Security Student Grant \$625	2024

Projects

Note: Descriptions in *Contribution* are confirmed by my collaborators.

CuSAFE sites.google.com/view/safe-gpu We built a system named CuSAFE. It detects memory safety vulnerabilities in CUDA programs. Our evaluation shows that CuSAFE beats existing tools such as compute-sanitizer (NVIDIA), cuCatch (PLDI '22, NVIDIA), and LMI (HPCA' 25) in both performance and accuracy; CuSAFE also supports the LLMs such as LLaMA2 and LLaMA3. CuSAFE is approx. $13\times$ faster than the compute-sanitizer, which is the official tool provided by NVIDIA . <i>Contribution:</i> I proposed this project and worked as the sole student; I completed the whole project myself.	2025
MOLE (CCS' 25) sites.google.com/view/mole-gpu We proposed a new attack against existing GPU TEEs; it leverages an under-documented MCU in Arm Mali GPUs to break the security of GPU TEEs. Arm have acknowledged our findings and enhanced their supply-chain security. It breaks multiple GPU TEEs, such as StrongBox (CCS' 22, Ant Group), CAGE (NDSS' 25, Ant Group) and My-TEE (NDSS' 23). <i>Contribution:</i> I proposed this project and worked as the leader with another student Yunjie Deng. I am responsible for reverse engineering and modifying core GPU MCU firmware, while Yunjie Deng analyzed the GPU programs and tested the attack prototype.	2025
MOAT (USENIX Security' 24) sites.google.com/view/safe-bpf We build a system named MOAT. It isolates BPF programs using hardware features like Intel MPK/Arm Stage-II. We have worked with a company to put it into production for 1,170,000 Yuan . <i>Contribution:</i> I proposed this project and worked as the leader; I completed the whole project myself. Collaborations with the company is supervised by me and developed by another student Lijian Huang.	2024
Patent ZL 2022 1 0436969.8 Kernel Space Debug Method, Device and Storage Medium	2022
Translation of Software Foundation coq-zh.github.io/SF-zh I helped translate the Software Foundation, a famous textbook about formal verification.	2022
BadUSB-C (WOOT' 21) We uncovered a novel attack vector against USB-C devices. It enhances the power of BadUSB attacks with the video capability of USB-C. We received 30,000 Yuan bounty award from the affected vendor.	2021